

Assignment 9 — Results Analysis and Interpretation

GP Practice Demographics and Hospital Service Use in Scotland

1. Introduction

This report synthesises the experimental results from Assignments 1–8, covering the full modelling cycle for the engineering thesis project. The central question is whether incorporating demographic variables — in particular the share of the GP-registered population aged 65 and over — significantly improves predictions of hospital service use across Scottish GP practices, relative to a baseline model that uses only practice size and GP availability. The following sections present consolidated comparative metrics, structured error and interpretability analysis, robustness diagnostics, and explicit alignment with the original research hypothesis.

A. Consolidated Comparative Results

Table 1 reports holdout and cross-validation performance for all six models evaluated across Assignments 7 and 8. The holdout partition consists of three observations from practice 1004 (Aberdeen City, NHS Grampian), held out as a full practice group to prevent data leakage. Cross-validation used three group-aware folds on the nine-observation training partition. All metrics are computed on the original scale of the target variable (hospital admissions per 1,000 registered patients).

Model	Features	R ²	RMSE	MAE	CV RMSE	CV σ (RMSE)	ΔR^2 vs OLS	Δ RMSE vs OLS
Dummy mean (F0)	F0	-818.568	52.6321	52.6	45.871	—	-801.951	-582.1%
OLS F0 (baseline)	F0	-16.6173	7.7166	7.5387	43.961	—	0.000	0.0% (ref)
Ridge F1 ($\alpha=0.1$, A7)	F1	-18.3574	8.0888	8.0828	8.314	± 3.672	-1.74	-4.8%
Decision Tree F1 (A7)	F1	-128.8218	20.9475	20.8667	57.627	—	-112.204	-171.5%
Ridge F1 ($\alpha=0.01$, A8)	F1	0.4685	1.3403	1.3353	4.8357	± 5.3228	17.086	82.6%
Random Forest F1 (A8)	F1	-297.0036	31.7372	31.6837	34.1703	—	-280.386	-311.3%

Primary model. The Ridge F1 ($\alpha=0.01$) achieves the only positive holdout R² (0.469) and the lowest RMSE (1.340) across all models. This represents an improvement of +17.1 pp in R² and an

82.6% reduction in RMSE relative to the OLS F0 baseline. Random Forest, despite being a more expressive model class, generalises poorly (holdout RMSE = 31.74), indicating that the sample size (n=12) is insufficient to support non-linear tree-based modelling. The Decision Tree baseline suffers the same limitation.

B. Structured Error Analysis

The test partition contains three observations from a single practice (1004, Aberdeen City). Although this limits the generalisability of error patterns, it enables precise examination of model behaviour on an unseen practice with characteristics that differ from the training set.

B.1 Test-Set Predictions and Residuals — Primary Model

Date	Actual (per 1,000)	Predicted	Residual (A-P)	Error
2024-01-01	248.6	250.016	-1.416	1.416
2024-02-01	246.2	247.617	-1.417	1.417
2024-03-01	244.1	245.273	-1.173	1.173

Mean residual: -1.335 | Std of residuals: 0.141 | Max absolute error: 1.417 per 1,000. The residuals show a small negative bias, suggesting the model slightly over-predicts hospital use for this practice. All three observations belong to the same practice (Aberdeen City), so the error pattern reflects a single structural deviation rather than random noise across diverse cases.

B.2 Comparison of Residual Magnitudes Across Models

The OLS F0 baseline produces residuals on the order of 7–8 units per 1,000, while the primary Ridge F1 model reduces this to approximately 1.3 units — a factor-of-six improvement. The Random Forest's residuals (around 32 units) far exceed those of all linear models, confirming severe overfitting to the nine-observation training set; its training error is near zero while its test error is catastrophic.

B.3 Structural Error Patterns and Limitations

Because the test set comprises a single practice across three consecutive months, observed residuals share a common structural component: any systematic misspecification for Aberdeen City will appear as a consistent bias rather than random error. Practice 1004 has the highest share of elderly population (20.5%) and the highest GP availability (0.73) among all four practices, yet exhibits the lowest hospital use (mean around 248 per 1,000). The model's low residuals suggest it captures this relationship adequately, but the evidence rests on three data points from one practice — a fundamental limitation acknowledged explicitly in Section E.

C. Model Interpretability and Feature Contribution

C.1 Ridge F1 Coefficient Analysis

Ridge regression with L2 regularisation shrinks coefficients towards zero but preserves their sign and relative magnitude. Table 3 presents the fitted coefficients from the primary model ($\alpha=0.01$, trained on standardised numeric features).

Feature	Coefficient (standardised)
intercept	+299.3095
log_all_ages	-0.0789
gp_availability	-11.0318
share_age_65_plus	-10.0536
share_female	+4.7813
deprivation_index	+5.6734
HB_NHS_Greater_Glasgow	+1.2285
HB_NHS_Lothian	-1.2285
HSCP_Edinburgh_North	-1.1286
HSCP_Edinburgh_West	-0.0999
HSCP_Glasgow_Central	+1.2285

Key interpretive findings:

1. Deprivation index (+5.67): The strongest positive predictor. Higher area deprivation is strongly associated with greater hospital admission rates, consistent with the established relationship between socioeconomic disadvantage and unplanned care demand in Scottish health literature.
2. GP availability (−11.03): Better access to primary care is associated with reduced hospital use. This aligns with the hypothesis that adequate GP supply reduces avoidable admissions through earlier intervention.
3. Share aged 65+ (−10.05): The demographic variable central to the research hypothesis shows a negative coefficient. This counter-intuitive finding warrants careful interpretation: in this dataset, practice 1004 (Aberdeen City) has the highest elderly share and the lowest hospital use, while practice 1003 (Glasgow Central) has the lowest elderly share and the highest deprivation-driven admissions. The negative sign likely reflects the confounding influence of deprivation and GP availability rather than a genuine protective effect of ageing. With four practices over three months, the model cannot isolate the demographic effect from correlated covariates.
4. Share female (+4.78): A modest positive association; differences in health-seeking behaviour may contribute, though coefficient magnitude is small relative to deprivation.
5. Practice size (log_all_ages, −0.08): Near-zero effect after controlling for other variables, confirming that raw population size is a weak predictor once demographics are included.
6. Health Board fixed effects: NHS Greater Glasgow and HSCP Glasgow Central carry a modest positive loading (+1.23), capturing regional factors not fully explained by deprivation and demographics.

C.2 Random Forest Feature Importances

For reference, Table 4 reports the impurity-based feature importances from the Random Forest benchmark. Despite catastrophic generalisation, the ranking provides a complementary view of feature relevance under a non-linear model.

Feature	Importance	% of total
deprivation_index	0.1899	19.0%
share_female	0.1754	17.5%
share_age_65_plus	0.1334	13.3%
log_all_ages	0.1268	12.7%
gp_availability	0.1153	11.5%
HB_NHS_Greater_Glasgow	0.0975	9.8%
HSCP_Glasgow_Central	0.0974	9.7%
HB_NHS_Lothian	0.0495	5.0%
HSCP_Edinburgh_North	0.0131	1.3%
HSCP_Edinburgh_West	0.0018	0.2%

The Random Forest ranks deprivation_index (19.0%), share_female (17.5%), and share_age_65_plus (13.3%) as the three most important features, broadly consistent with the Ridge coefficient magnitudes. The broad distribution of importance across all features suggests that no single variable dominates; predictive power arises from the joint demographic-clinical profile of each practice.

D. Robustness and Stability Assessment

D.1 Alpha Sensitivity — Ridge F1 Hyperparameter Grid

Assignment 8 expanded the regularisation search from three values ($\alpha \in \{0.1, 1.0, 10.0\}$) to seven orders of magnitude. Table 5 reports CV mean RMSE and its standard deviation across folds for each α value.

Alpha	CV Mean RMSE	CV σ (RMSE)	CV Mean R^2	
0.001	9.4396	± 4.8869	-28.9057	
0.01	4.8357	± 5.3228	-6.6730	← best
0.1	8.3142	± 3.6716	-38.2305	
1.0	14.8238	± 6.7899	-122.6510	
10.0	14.3538	± 6.1308	-161.3466	
100.0	30.2656	± 18.7387	-580.0484	
1000.0	43.8502	± 23.1068	-970.3960	

The optimal value $\alpha=0.01$ sits at the lower end of the grid, implying minimal regularisation. CV RMSE rises steeply on both sides, indicating a narrow optimal region. The standard deviation of CV RMSE at $\alpha=0.01$ is ± 5.3228 , which is large relative to the mean (4.836), revealing high fold-to-fold variability. This instability is expected given that each fold contains only two or three training observations from one or two practices.

D.2 Generalisation Gap — Train vs Cross-Validation RMSE

Table 6 quantifies the gap between training RMSE (fitted on the full training set) and CV RMSE, which approximates generalisation error.

Alpha	Train RMSE	CV RMSE	CV σ	Gap (CV–Train)
0.001	0.2440	9.4396	± 4.8869	9.1956
0.01	0.3852	4.8357	± 5.3228	4.4505
0.1	0.5586	8.3142	± 3.6716	7.7556
1.0	1.0708	14.8238	± 6.7899	13.7530
10.0	6.2937	14.3538	± 6.1308	8.0601
100.0	22.9161	30.2656	± 18.7387	7.3495
1000.0	31.8051	43.8502	± 23.1068	12.0451

At $\alpha=0.01$ (the selected value), training RMSE = 0.3852 against a CV RMSE of 4.8357, yielding a gap of 4.4505. Although lower α values (0.001) achieve near-zero training RMSE, their CV RMSE is larger, confirming overfitting at very low regularisation. The selected model therefore represents the best available trade-off on this dataset, but the gap remains substantial in absolute terms.

D.3 Robustness Summary

The primary model's holdout performance (RMSE=1.340) appears strong but must be interpreted with caution for three reasons: (i) the test set contains a single practice observed over three consecutive months, meaning temporal autocorrelation inflates apparent accuracy; (ii) cross-validation RMSE (4.84) is 3.6 $\times$ larger than holdout RMSE, indicating that the holdout result may not represent typical out-of-sample performance; (iii) the high CV fold-to-fold standard deviation suggests that conclusions are sensitive to which practice is held out in each fold. These limitations are structural and inherent to the twelve-observation sample — they do not invalidate the experimental findings but bound the confidence with which they can be generalised.

E. Alignment with Research Question and Hypothesis

E.1 Research Question

Research question (Assignment 1): Does incorporating demographic structure of GP practice populations — particularly the share of patients aged 65 and over — improve predictions of hospital service use beyond a baseline model using only practice size and GP availability? The success criteria defined in Assignment 1 were: $\Delta R^2 \geq 5$ percentage points and $\Delta \text{RMSE} \geq 5\%$ relative to OLS F0.

E.2 Empirical Verdict

On the holdout set: $\Delta R^2 = +17.086$ pp (threshold: $\geq +0.050$) and $\Delta \text{RMSE} = 82.6\%$ (threshold: $\geq 5\%$). Both criteria are exceeded by large margins. The hypothesis that demographic variables improve hospital use prediction is supported empirically on this dataset.

E.3 Qualifications and Limitations

1. Sample size: The dataset contains 12 observations from 4 practices over 3 months. Statistical inference is not reliable at this scale; the success criteria are met numerically but the results are not generalisable beyond this sample without validation on the full Public Health Scotland data.
2. Single-practice test set: All holdout observations come from one practice (1004, Aberdeen City). The strong performance may reflect characteristics specific to that practice rather than the general relationship between demographics and hospital use.
3. Coefficient sign: The negative coefficient on `share_age_65_plus` (-10.05) conflicts with the directional hypothesis (older populations $\rightarrow$ higher hospital use). This finding likely reflects confounding with deprivation and GP availability in a four-practice dataset, and should not be interpreted as evidence that older populations reduce hospital admissions. Causal claims are not supported by this analysis.
4. Temporal structure: Repeated measures from the same practice over three months introduce autocorrelation that is not accounted for in the model. This inflates apparent predictive accuracy.
5. Nonlinear benchmark failure: The Random Forest benchmark ($RMSE=31.74$) failed dramatically, which is expected for complex models on small samples. This confirms that the linear Ridge model is the appropriate class for this data regime.

E.4 Conclusion

The experimental evidence supports the hypothesis within the scope of the available data. Adding demographic variables — particularly deprivation index, GP availability, elderly share, and geographic Health Board fixed effects — yields a model (Ridge F1, $\alpha=0.01$) with holdout $R^2=+0.4685$ and $RMSE=1.3403$, compared to $R^2=-16.6173$ and $RMSE=7.7166$ for the demographic-free baseline. The improvement is substantial and consistent across both primary metrics. However, the scale of this sample requires that these conclusions be treated as preliminary: the analysis demonstrates methodological feasibility and directional evidence rather than statistical certainty.